

Umar Bin Moiz

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PROFESSIONAL SUMMARY

AI Engineer with 2.5+ years building production-grade agentic AI systems and RAG infrastructure in regulated financial services environments. Hands-on experience designing and deploying multi-agent pipelines, MCP-aligned agent frameworks, and internal knowledge-base engines powered by RAG architecture. Proven ability to collaborate cross-functionally to identify AI enablement opportunities and translate vague business requirements into clear, executable technical plans. MSc in Digital Financial Technology, NUS (2026).

CORE COMPETENCIES

Programming Skills/Languages: Python, Java, JavaScript, TypeScript, C++, Object Oriented Programming(OOP)

AI/ML Frameworks & Libraries: PyTorch, Deep Learning, scikit-learn, XGBoost, LightGBM, Random Forests, LangGraph, Langchain, vLLM, ONNX, FastT5, Transformers, Hugging Face

Generative AI & NLP: Large Language Models (LLMs), Prompt Engineering, LLM Fine-tuning & Evaluation, RLHF, Agentic/RAG Systems, Vector Databases

ML Platform & Infrastructure: Kubernetes, OpenShift, Docker, MLOps, Model Serving/Quantization, Model Monitoring, GPU Optimization (V100/A100/H100), CI/CD (Jenkins, GitHub Actions)

Model Lifecycle & Governance: Model Registry, Drift Detection, Feature Stores, Rollback Strategies, Responsible AI, Guardrails

Cloud & Data Processing: AWS, Google Cloud, Redis, FastAPI, RESTful APIs

Databases: PostgreSQL, MongoDB, Pinecone, SQL

Monitoring & Observability: Grafana, Logging Systems, Model Monitoring, Telemetry, Incident Triage

Development Tools: Git, Unix,Excel, MERN Stack (MongoDB, Express, React, Node.js)

SOFT SKILLS

Leadership & Collaboration: Cross-functional team leadership, mentoring junior developers, stakeholder management

Communication: Technical documentation, presenting complex AI concepts to non-technical audiences, code reviews

Problem Solving: System design, troubleshooting production issues, performance optimization

Project Management: Agile/Scrum methodologies, sprint planning, delivering projects on tight deadlines

WORK EXPERIENCE

Artificial Intelligence Research and Development Intern

June 2025 – November 2025

Marsh McLennan

Singapore

- Architected and developed “Genie Broker”, an end-to-end LLM-powered insurance renewal automation platform using LangGraph multi-agent orchestration, RAG-based document extraction, and FastAPI/TypeScript; led proof-of-concept development through to production-ready design, projecting 75–90% reduction in manual processing time; created comprehensive technical documentation and handover materials enabling seamless knowledge transfer to the development team
- Researched and evaluated agentic AI tools (AutoGen, CrewAI, LangChain, LangGraph) across determinism, reliability, and production-readiness criteria; selected LangGraph for enterprise deployment
- Configured CI/CD pipeline via Polaris with Docker containerization in an enterprise environment
- Proactively engaged business teams to uncover AI enablement opportunities across 4 policy complexity tiers; translated vague requirements into clear, executable technical roadmaps and agent-powered workflow solutions
- Mentored other interns on agentic AI development practices, LangGraph implementation, and enterprise deployment standards; created technical documentation and guides enabling developers to independently understand and apply agentic AI concepts

ML Ops & Software Engineer, Conversational AI

July 2022 – August 2024

Development Bank of Singapore (DBS)

Singapore

- Developed GenAI virtual assistant with RAG-based knowledge retrieval and real-time transcription, enabling live search during calls and automated post-call documentation for 500 Customer Service Officers serving 250,000+ monthly customers; reduced manual review time by 90% and overall call handling time by 20%, enhancing customer experience through faster resolution and improved agent efficiency

- Presented GenAI virtual assistant demo to senior leadership and cross-functional stakeholders — translating complex LLM inference capabilities, architectural trade-offs, and deployment constraints into clear business value; directly influenced the bank’s strategic decision to expand GenAI initiatives bank-wide
- Built and optimized production LLM serving infrastructure using vLLM, ONNX, and FastT5 with quantization and batch processing on A100/H100 GPU clusters, capable of scaling to 6,000+ concurrent users with low-latency inference; monitored production uptime via telemetry and observability dashboards (Grafana), supporting incident triage and minimizing downtime
- Managed end-to-end OpenShift (Kubernetes) deployments for Conversational AI platform including LLM models, APIs, and services, implementing auto-scaling and resource optimization that enabled efficient GPU utilization and zero-downtime updates, managed model versioning and selection via internal model registry, with rollback strategies to maintain production stability
- Engineered vector database infrastructure and developed prompt engineering strategies for RAG-based NLP Q&A systems, evaluating multiple LLM configurations before production deployment — improving retrieval speed by 70%
- Built LLM-powered web crawler and data extraction tool using Google Cloud/Vertex AI for automated Know Your Customer(KYC) verification, reducing manual verification time by 80% and accelerating compliance workflows

Data Analyst

December 2020 – July 2021

DNV

Singapore

- Developed internal automation platform using MERN stack (MongoDB, Express, React, Node.js), improving operational efficiency through streamlined data processing and workflow automation
- Implemented machine learning solution for vessel fuel efficiency optimization using AIS data and predictive modeling to determine optimal speed over ground, reducing fuel consumption; integrated model outputs into a full-stack dashboard via Power BI and custom frontend for real-time operational decision making

EDUCATION

National University of Singapore (NUS)

Singapore

Master of Science in Digital Financial Technology

August 2024 - June 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning for Finance, Risk Analytics for Financial Services, Financial Modelling, Fintech Innovation, Deep Learning for Finance, Neural Networks
- **Projects:** Developed predictive models using classical ML algorithms (XGBoost, Random Forests) and deep learning for stock market prediction and financial forecasting; evaluated model performance and feature engineering techniques on time-series financial data

National University of Singapore (NUS)

Singapore

Bachelor of Engineering in Computer Engineering

August 2018 - June 2022

- **Relevant Coursework:** Software Engineering, Artificial Intelligence, Computer Vision, Natural Language Processing, Machine Learning, Blockchain Engineering

ACHIEVEMENTS

Varsity Sailing: NUS Sailing Team — represented NUS at World Championships (2025); competed in Grand Slam regattas (2019, 2020, 2025, 2026)

CERTIFICATIONS

AWS Certified Solutions Architect - Associate | AWS Certified Cloud Practitioner | Google Cloud Certified - Associate Cloud Engineer